see any plausible way for nuclear energy to power quasars. It was still too puny.

So the only possibility left was *gravitational energy*, the same kind of energy we were driven to, when navigating the *Endurance* around Gargantua. For the *Endurance*, gravitational energy was harnessed by a slingshot around an intermediate-mass black hole (Chapter 7). The black hole's intense gravity was key. For quasars, similarly, the power must come from a black hole.

For several years, astrophysicists struggled to figure out how a black hole could do the job. The answer was found in 1969, by Donald Lynden-Bell at the Royal Greenwich Observatory in England. A quasar, Lynden-Bell hypothesized, is a gigantic black hole surrounded by a disk of hot gas (an accretion disk) that is threaded by a magnetic field (Figure 9.2).

Hot gas in our universe is almost always threaded by magnetic fields (Chapter 2). These fields are locked into the gas; the gas and fields move together, in lockstep.

When threading an accretion disk, a magnetic field becomes a catalyst for converting gravitational energy into heat and then light. The field provides ultrastrong friction[21] that slows the gas's circumferential motion, reducing the centrifugal force that holds it out against the pull of gravity, so the gas moves inward, toward the black hole. As the gas moves inward, the hole's gravity speeds up its orbital motion by even more than the friction slowed it. In other words, gravitational energy is converted into kinetic energy (energy of motion). Magnetic friction then converts half that new kinetic energy into heat and light, and the process repeats.

The energy comes from the black hole's gravity. The agents for extracting it are magnetic friction and the disk's gas.